

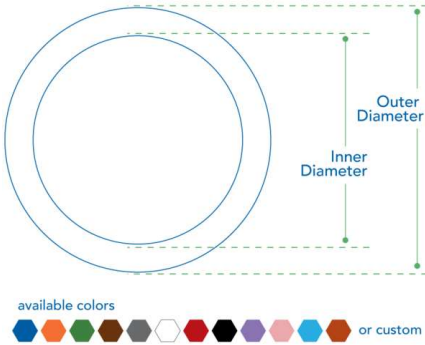
Exhibit B

Material Project Specification Sheets

WE DO THAT ... **& MORE**

SMOOTHWALL

1 1/4" SDR 11



Nominal OD	1.66 in
OD Tolerance +/-	0.008 in
Min. Wall	0.151 in
Wall Tolerance +	0.02 in
Min. ID	1.318 in
Avg ID	1.338 in
Weight	0.312 lb/ft
Bend Radius Sup	17 in
Bend Radius UnSup	34 in
Conduit SWPS	1717 lbs

STANDARD DETAILS

DETAILS	Manufactured from flexible HDPE (High Density Polyethylene)
SPECIFICATIONS	All Smoothwall conduit dimensions meet or exceed one or more of the following: ASTM F-2160, ASTM D-3350, ASTM D-3485, NEMA TC-7, UL 651A, UL 1990, Bellcore GR-356, CSA C22.2 No. 327-18
INSTALLATION TYPES	Plow, Trench, Directional Bore, Direct Burial, Tray Installations, Subdivided Conduit, Aerial
COLORS	Blue, Orange, Green, Brown, Grey, White, Red, Black, Lilac, Pink, Aqua, Terra Cotta, Lime Green, or custom colors with optional stripes
CONDUIT MARKINGS	Permanent marking along conduit includes: material, relevant standards, production info, and sequential feet or meter markings. Custom options available.

OPTIONS

CO-EXTRUDED LINING	SILICORE® ULF (Ultra-Low Friction) is co-extruded inside the HDPE wall creating a slick, permanent, interior lining. With a coefficient of friction 60% lower than standard HDPE conduit without the aid of wet lubricants, SILICORE® ULF exhibits no loss in performance over time or in extreme temperature conditions. Coefficient of friction for SILICORE ULF 0.07 versus non-lubricated HDPE conduit 0.18-0.22 (internal test method).
PRE-INSTALLED TAPE	Factory pre-installed Bull-Line™ Pull Tape with EVEN-LOAD™ ensures extra slack at any access point throughout the reel. Available 500lb–6,000lb tensile strength or locatable.
PRE-INSTALLED CABLE	Cable can be factory pre-installed in conduit
UV PROTECTANT	Available for UV exposure applications (Aerial, Lashed, or External Tray)

REEL SPECIFICATIONS

REEL SIZE	STANDARD LENGTH	REEL TYPE
50 x 46 in	1000 ft	Steel
72 x 46 in	4000 ft	Steel
96 x 46 in	8000 ft	Steel

† Safe working pull strength is calculated at 80% of tensile or breaking strength

* Unsupported Bend Radius guidelines should be followed during the installation process. The Supported Bend Radius are post-installation measurements.

PRODUCT COMPLIANCE

We offer an extensive portfolio of US-made conduit products and accessories that conform with several domestic preference standards, including Buy America (BAA), Build America, Buy America (BABA), and DOT Buy American requirements. Any products including a locate wire or preinstalled cables, will need to be specifically evaluated. Fire Retardant Resins may also require additional evaluation to meet program guidelines. Please contact your Dura-Line representative for more information.



HDPE CONDUIT TECHNICAL DATA SHEET

SMOOTHWALL RECOMMENDED ACCESSORIES

1 1/4" SDR 11

1 1/4" SHUR-LOCK II		1 1/4" CLEAR-LOCK		ROTARY CONDUIT CUTTER 0.75" TO 2.625"	
Quantity: 10 / Per Case Packaging: Case Part Number: 20000120		Quantity: 50 / Per Case Packaging: Each Part Number: 20005096		Quantity: 1 Packaging: Each Part Number: 20001924	
RATCHET CUTTER FOR 0.75" - 1.25" OD CONDUIT		KWIK CUTTER - CUTS UP TO 2.00" OD CONDUIT		CONDUIT SLITTER	
Quantity: 1 Packaging: Each Part Number: 20001923		Quantity: 1 Packaging: Each Part Number: 20001922		Quantity: 1 Packaging: Each Part Number: 20005686	
1.25" SDR FLAT VINYL END CAP 1.660" OD		1.25" SDR TAPERED VINYL END CAP 1.640" OD		1.25" EXPANSION PLUG 1.156" - 1.510" ID	
Quantity: 1 Packaging: Each Part Number: 20001738		Quantity: 1 Packaging: Each Part Number: 20001739		Quantity: 50 / Per Case Packaging: Each Part Number: 20001871	

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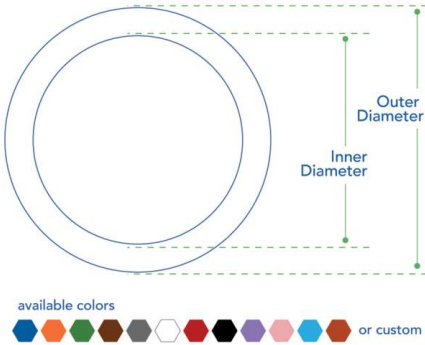
HYDRALUBE F100I SUMMER (12) 1 QUART		HYDRALUBE F150I SUMMER (4) 1 GALLON		BULL-LINE BUCKET WP12 2100FT	
Quantity: 24 / Per Pallet Packaging: (12) 1 Quart Part Number: 20000209		Quantity: 30 / Per Pallet Packaging: (4) 1 Gallon Part Number: 20000211		Quantity: 36 Packaging: N/A Part Number: 20095169	

BULL-LINE BUCKET WP18 1300FT	
Quantity: 36 Packaging: N/A Part Number: 20095170	

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SMOOTHWALL

6" SDR 11



Nominal OD	6.625 in
OD Tolerance +/-	0.033 in
Min. Wall	0.602 in
Wall Tolerance +	0.072 in
Min. ID	5.277 in
Avg ID	5.349 in
Weight	4.944 lb/ft
Bend Radius Sup	73 in
Bend Radius UnSup	146 in
Conduit SWPS	27349 lbs

STANDARD DETAILS

SPECIFICATIONS	All Smoothwall conduit dimensions meet or exceed one or more of the following: ASTM F-2160, ASTM D-3350, ASTM D-3485, NEMA TC-7, UL 651A, UL 1990, Bellcore GR-356, CSA C22.2 No. 327-18
INSTALLATION TYPES	Plow, Trench, Directional Bore, Direct Burial, Tray Installations, Aerial
COLORS	Blue, Orange, Green, Brown, Grey, White, Red, Black, Lilac, Pink, Aqua, Terra Cotta, Lime Green, or custom colors with optional stripes
CONDUIT MARKINGS	Permanent marking along conduit includes: material, relevant standards, production info, and sequential feet or meter markings. Custom options available.

OPTIONS

CO-EXTRUDED LINING	SILICORE® ULF (Ultra-Low Friction) is co-extruded inside the HDPE wall creating a slick, permanent, interior lining. With a coefficient of friction 60% lower than standard HDPE conduit without the aid of wet lubricants, SILICORE® ULF exhibits no loss in performance over time or in extreme temperature conditions. Coefficient of friction for SILICORE ULF 0.07 versus non-lubricated HDPE conduit 0.18-0.22 (internal test method).
PRE-INSTALLED TAPE	Factory pre-installed Bull-Line™ Pull Tape with EVEN-LOAD™ ensures extra slack at any access point throughout the reel. Available 500lb–6,000lb tensile strength or locatable.
PRE-INSTALLED CABLE	Cable can be factory pre-installed in conduit up to 3"
UV PROTECTANT	Available for UV exposure applications (Aerial, Lashed, or External Tray)

REEL SPECIFICATIONS

REEL SIZE	STANDARD LENGTH	REEL TYPE
120 x 48 in	450 ft	Steel
-	20 ft	Stick
-	40 ft	Stick

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PRODUCT COMPLIANCE

We offer an extensive portfolio of US-made conduit products and accessories that conform with several domestic preference standards, including Buy America (BAA), Build America, Buy America (BABA), and DOT Buy American requirements. Any products including a locate wire or preinstalled cables, will need to be specifically evaluated. Fire Retardant Resins may also require additional evaluation to meet program guidelines. Please contact your Dura-Line representative for more information.



HDPE CONDUIT TECHNICAL DATA SHEET

SMOOTHWALL RECOMMENDED ACCESSORIES

6" SDR 11

6" SHUR-LOCK II		6" REDI-LOCK		6.00" SDR E-LOC 6" LENGTH	
Quantity: 2 / Per Case Packaging: Case Part Number: 20000256		Quantity: 2 / Per Case Packaging: Case Part Number: 20005482		Quantity: 1 Packaging: Each Part Number: 20001503	
6.00" EXPANSION PLUG 5.975" - 6.375" ID		HYDRALUBE F200I SUMMER (4) 1 GALLON		HYDRALUBE F300I SUMMER 5 GALLON PAIL	
Quantity: 20 / Per Case Packaging: Each Part Number: 20001763		Quantity: 30 / Per Pallet Packaging: (4) 1 Gallon Part Number: 20000218		Quantity: 24 / Per Pallet Packaging: 5 Gallon Drum Part Number: 20000005	
LUBADUK SUMMER (6) 1 GALLON		BULL-LINE BUCKET WP25 750FT		BULL-LINE BUCKET WP18 1300FT	
Quantity: 18 / Per Pallet Packaging: (6) 1 Gallon Part Number: 20000351		Quantity: 36 Packaging: N/A Part Number: 20095171		Quantity: 36 Packaging: N/A Part Number: 20095170	

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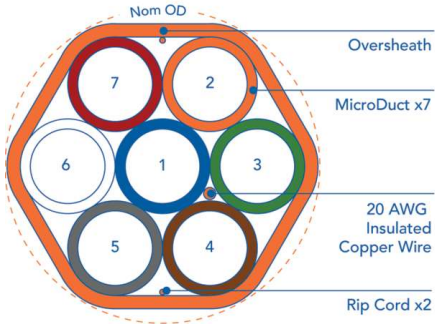


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FUTUREPATH

7-way 18/14 mm



MicroDuct OD/ID	18/14 mm
Nominal OD	2.27 in
MicroDuct Min ID	14 mm
Oversheath	0.07 in
Weight	0.656 lb/ft
Bend Radius Sup	31 in
Bend Radius UnSup	52 in
Conduit SWPS	3522 lbs

STANDARD DETAILS

DETAILS	FuturePath is a unit of bundled MicroDucts. Manufactured from flexible HDPE (High Density Polyethylene). FuturePath dimensions and performance meet or exceed the requirements of Telcordia GR-3155 and are compliant with ASTM D3350.
INSTALLATION TYPES	Subdivided Conduit, Overrides, Plow, Trench, Directional Bore, MicroTrench, Tray
FILL RATIO	Choose the correct MicroDuct size based on the Outer Diameter (OD) of desired MicroCable. Dura-Line recommends a fill ratio of 50% to 75% for optimal cable placement performance. Several factors impact jetting distance including the condition of route, bends, and equipment.
COLORS	Oversheath: Orange MicroDucts: (1) Blue, (2) Orange, (3) Green, (4) Brown, (5) Grey, (6) White, (7) Red
CONDUIT MARKINGS	Permanent marking along FuturePath includes: material, relevant standards, production info, and sequential feet or meter markings. Custom options available.
CO-EXTRUDED LINING	SILICORE® ULF (Ultra-Low Friction) is co-extruded inside the HDPE wall creating a slick, permanent, interior lining. SILICORE® ULF exhibits no loss in performance over time or in extreme temperature conditions.
INTERNAL RIBS	Standard (except 3.5mm ID MicroDucts which are designed with a standard smooth interior)
LOCATE WIRE	Includes a 20 AWG insulated copper wire
RIP CORD(S)	For easy opening of the oversheath
OPTIONS	
THICKER OVERSHEATH	Available in most configurations to meet your needs for more rugged projects

REEL SPECIFICATIONS

REEL SIZE	STANDARD LENGTH	REEL TYPE
96 x 46 in	4000 ft	Steel

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PRODUCT COMPLIANCE

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FUTUREPATH RECOMMENDED ACCESSORIES

7-way 18/14 mm

18MM STRAIGHT COUPLER 18/14MM (OD/ID)		18MM END CAP		LONGITUDINAL SHEATH SLITTER	
Quantity: 1 Packaging: Each Part Number: 20001518		Quantity: 1 Packaging: Each Part Number: 20001485		Quantity: 1 Packaging: Each Part Number: 20003768	
MICRODUCT ROUND CUTTER		LARGE MICRODUCT CUTTER - UP TO 28MM OD		18MM GAS BLOCK CONNECTOR (7.0-10.0MM CABLE)	
Quantity: 1 Packaging: Each Part Number: 20005284		Quantity: 1 Packaging: Each Part Number: 20005684		Quantity: 1 Packaging: Each Part Number: 20004052	
18MM DIVISIBLE DUCT SEAL CABLE OD 3.50MM-12.00MM		HYDRALUBE AT-500 SUMMER (4) 1 GALLON		7-WAY 18MM SPLICE KIT	
Quantity: 10 / Per Case Packaging: Case Part Number: 20005189		Quantity: 30 / Per Pallet Packaging: (4) 1 Gallon Part Number: 20000129		Quantity: 1 Packaging: Each Part Number: 20001533	
210 LB BULLET-LINE PAIL 6500'		FUTUREPATH ENTRY SEAL 7-WAY 16MM & 18/14			
Quantity: 1 Packaging: Each Part Number: 20000282		Quantity: 1 Packaging: Each Part Number: 20095076			

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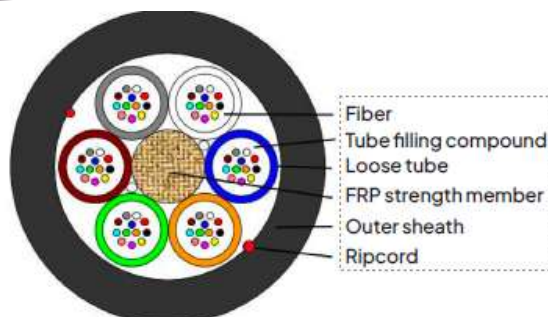
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Micro-Fiber (200μm)

12F, 24F, 48F, 72F, 96F, 144F, 192F, 216F, 288F, 432F, 576F



Typical Construction



General/Scope

This Specification covers the design requirements and performance standard for the supply of optical fiber cable in the industry. RaDD ensures a stable quality control system for our cable products through several programs including ISO 9001, ISO 14001 and OHS.

Cable type

Micro-Fiber Cable (200μm)

Application

Suitable for air blowing installation

Reference

ITU-T G.657	Characteristics of a single-mode optical fiber
IEC 60794-1-1	Optical fiber cables- part 1-1-Generic specification - General
IEC 60794-1-21	Optical fiber cables - part 1-2-Generic specification - Basic optical cable test procedure - Mechanical test methods
IEC 60794-1-22	Optical fiber cables - part 1-2-Generic specification - Basic optical cable test procedure - Environmental test methods
IEC 60794-3	Optical fiber cables- part 3 - Sectional specification - Outdoor cables
IEC 60794-5-10	Optical fiber cables - Part 5-10 Family specification for outdoor micro-duct optical and protected micro-ducts for installation by blowing

Lifetime

Optical fiber cables supplied in compliance with the specifications can be capable of withstanding the typical service condition for a period of twenty-five (25) years without detriment to the transmission or operation and maintenance characteristics of the cable.

CONTACT & INFORMATION:

Tel: (920) 328-1020 Email: info@raddnetwork.com Website: RaDDNetwork.com



Application

Item	Value
Operation temperature	-22°F ~ +158°F
Installation temperature	14°F ~ +122°F
Storage temperature	-22°F ~ +158°F
Static bending radius	10 times the cable diameter
Dynamic bending radius	20 times the cable diameter

Optical Fiber

Optical Fibers supplied in this specification meet the requirements of ITU-G657A1 (200μm).

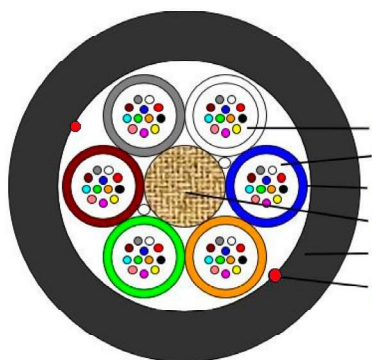
Category	Description	Specification
Geometrical Characteristics	Cladding diameter	125.0 ± 0.7 μm
	Cladding non-circularity	≤ 0.7 %
	Core concentricity error	≤ 0.5 μm
	Coating diameter	190~210 μm (Before Colored) 195 ± 225 μm (Colored)
	Coating/cladding concentricity error	≤ 12 μm
Optical Characteristics	Mode field diameter at 1310 nm	8.4 ± 9.2 μm
	Attenuation at 1310 nm	≤ 0.36 dB
	Attenuation at 1550 nm	≤ 0.22 dB/km
	Point discontinuity at 1310 nm and 1550 nm	≤ 0.05 dB
	Zero dispersion wavelength	1300 ~ 1324 nm
	Zero dispersion slope	≤ 0.092 ps/(nm ² ·km)
	Cable cut-off wavelength (λ _{cc})	≤ 1260 nm
	Polarization mode dispersion individual fiber	≤ 0.2 ps/√km
	Polarization mode dispersion design link value (M = 20, Q = 0.01%)	≤ 0.1 ps/√km
	Macro-bend loss (10 turns, 15 mm radius)	1550nm: ≤ 0.25 dB; 1625nm: ≤ 1.0 dB;
	Macro-bend loss (1 turn, 10 mm radius)	1550nm: ≤ 0.75 dB; 1625nm: ≤ 1.5 dB;
Mechanical Specification	Proof stress level	≥ 100kpsi (0.69 GPa)
	Coating strip force (peak value)	1.3~8.9 N
	Dynamic Fatigue Parameter (nd)	≥ 20
	Fiber curl (Radius)	≥ 2m

Optical Cable

Technical Characteristics

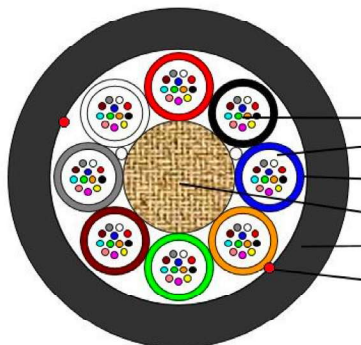
- The unique second coating and stranding technology provide the fibers with enough space and bending endurance, which ensure good optical property of the fibers in the cable
- Accurate process control ensures good mechanical and temperature performance
- High quality raw material guarantees the long service life of cable

Cross Section of Cable



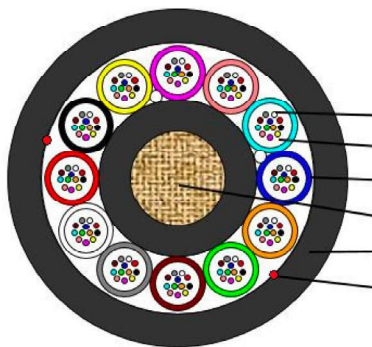
12F~72F

Fiber
Tube filling compound
Loose tube
FRP strength member
Outer sheath
Ripcord



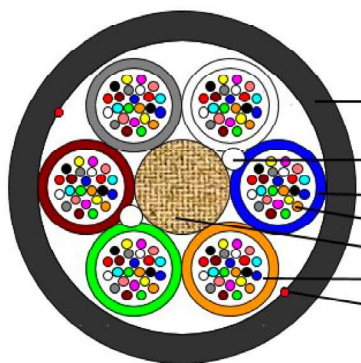
96F

Fiber
Tube filling compound
Loose tube
FRP strength member
Outer sheath
Ripcord



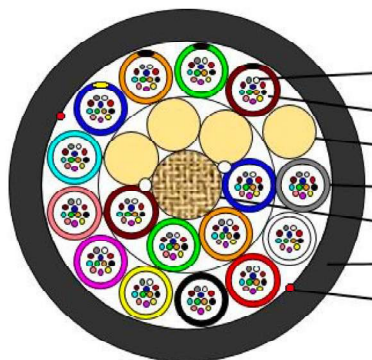
144F

Fiber
Tube filling compound
Loose tube
FRP strength member
Outer sheath
Ripcord



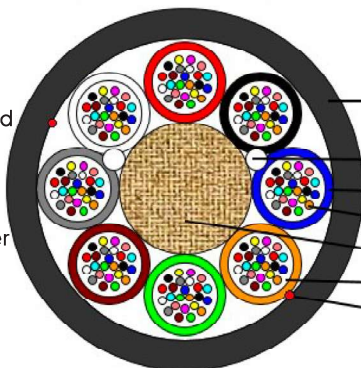
144F

Outer sheath
Water blocking yarn
Loose tube
Fiber
FRP strength member
Tube filling compound
Ripcord



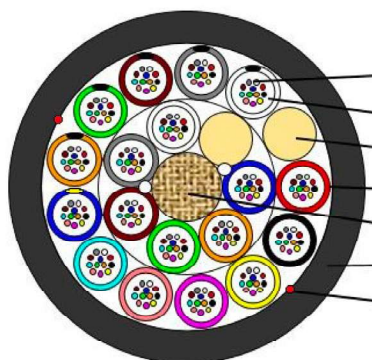
192F

Fiber
Tube filling compound
Filler
Loose tube
FRP strength member
Outer sheath
Ripcord



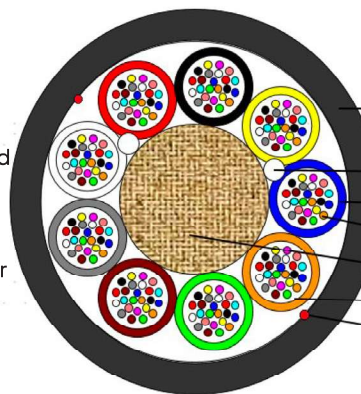
192F

Outer sheath
Water blocking yarn
Loose tube
Fiber
FRP strength member
Tube filling compound
Ripcord



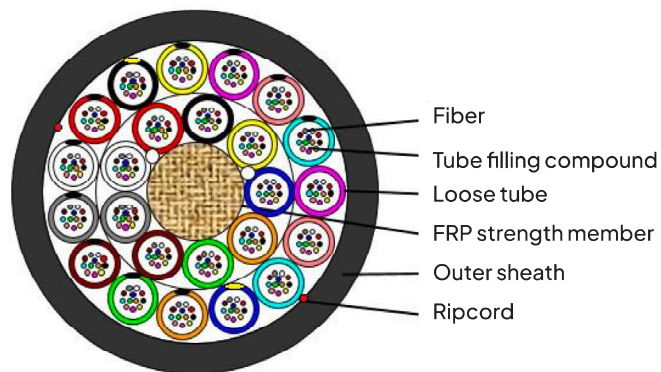
216F

Fiber
Tube filling compound
Filler
Loose tube
FRP strength member
Outer sheath
Ripcord

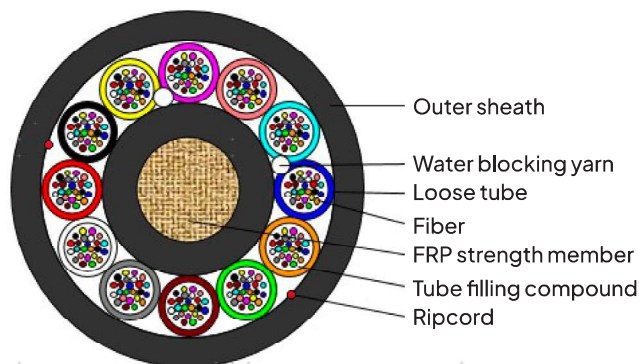


216F

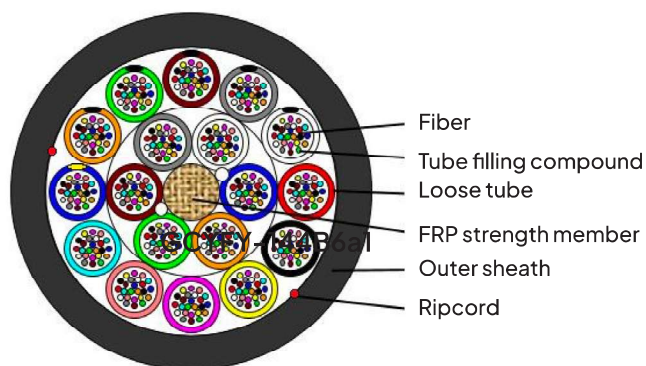
Outer sheath
Water blocking yarn
Loose tube
Fiber
FRP strength member
Tube filling compound
Ripcord



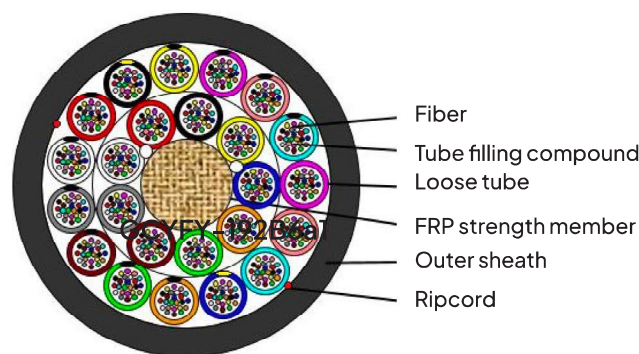
GCYFY-288B6a1 200um 24*12



GCYFY-288B6a1 200um 12*24



GCYFY-432B6a1 200um 18*24



GCYFY-576B6a1 200um 24*24

Structure of other fiber counts refer to **Dimensions and Descriptions of Cable Constructions**.
Schematic is for reference only.

Fiber and Loose Tube Identification

The color code of fibers and loose tube will be identification in accordance with the following color sequence, other sequence also is available. The color of the fillers will be natural.

Fiber Color Code						
6 fibers per tube	1	2	3	4	5	6
	Blue	Orange	Green	Brown	Slate	White
12 fibers per tube	1	2	3	4	5	6
	Blue	Orange	Green	Brown	Slate	White
	7	8	9	10	11	12
24 fibers per tube	Red	Black	Yellow	Purple	Pink	Aqua
	1	2	3	4	5	6
	Blue	Orange	Green	Brown	Slate	White
	7	8	9	10	11	12
	Red	Black	Yellow	Purple	Pink	Aqua
	13	14	15	16	17	18
	Blue with black ring	Orange with black ring	Green with black ring	Brown with black ring	Slate with black ring	White with black ring
	19	20	21	22	23	24
	Red with black ring	Natural	Yellow with black ring	Purple with black ring	Pink with black ring	Aqua with black ring

CONTACT & INFORMATION:

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Tube Color Code						
6~12 tubes	1	2	3	4	5	6
	Blue	Orange	Green	Brown	Slate	White
	7	8	9	10	11	12
18 tubes	Red	Black	Yellow	Purple	Pink	Aqua
	Inner 1	Inner 2	Inner 3	Inner 4	Inner 5	Inner 6
	Blue	Orange	Green	Brown	Slate	White
	Outer 1	Outer 2	Outer 3	Outer 4	Outer 5	Outer 6
	Red	Black	Yellow	Purple	Pink	Aqua
	Outer 7	Outer 8	Outer 9	Outer 10	Outer 11	Outer 12
16 tubes + 4 Fillers	Blue with yellow stripe	Orange with black stripe	Green with black stripe	Brown with black stripe	Slate with black stripe	White with black stripe
	Inner 1	Inner 2	Inner 3	Inner 4	Inner 5	Inner 6
	Blue	Orange	Green	Brown	Slate	Filler
	Inner 7	Outer 1	Outer 2	Outer 3	Outer 4	Outer 5
	Filler	Slate	White	Red	Black	Yellow
	Outer 6	Outer 7	Outer 8	Outer 9	Outer 10	Outer 11
	Purple	Pink	Aqua	Blue with yellow stripe	Orange with black stripe	Green with black stripe
	Outer 12	Outer 13				
18 tubes + 2 Fillers	Brown with black stripe	Filler				
	Inner 1	Inner 2	Inner 3	Inner 4	Inner 5	Inner 6
	Blue	Orange	Green	Brown	Slate	White
	Inner 7	Outer 1	Outer 2	Outer 3	Outer 4	Outer 5
	Filler	Red	Black	Yellow	Purple	Pink
	Outer 6	Outer 7	Outer 8	Outer 9	Outer 10	Outer 11
	Aqua	Blue with yellow stripe	Orange with black stripe	Green with black stripe	Brown with black stripe	Slate with black stripe
	Outer 12	Outer 13				
24 tubes	White with black stripe	Filler				
	Inner 1	Inner 2	Inner 3	Inner 4	Inner 5	Inner 6
	Blue	Orange	Green	Brown	Slate	White
	Inner 7	Inner 8	Inner 9	Outer 1	Outer 2	Outer 3
	Red	Black	Yellow	Purple	Pink	Aqua
	Outer 4	Outer 5	Outer 6	Outer 7	Outer 8	Outer 9
	Blue with yellow stripe	Orange with black stripe	Green with black stripe	Brown with black stripe	Slate with black stripe	White with black stripe
	Outer 10	Outer 11	Outer 12	Outer 13	Outer 14	Outer 15
	Red with black stripe	Black with yellow stripe	Yellow with black stripe	Purple with black stripe	Pink with black stripe	Aqua with black stripe

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Dimensions and Descriptions of Cable Construction

Item	Contents	Value																		
		12	24	36	24	48	72	96	144	192	216	288	96	144	192	216	288	432	576	
Filler	Number	2	4	6	2	4	6	8	12	16	18	24	4	6	8	9	12	18	24	
	Outer diameter (±0.1mm)	1.0			1.2								1.7							
	Number	4	2	0	4	2	0	0	0	3+1	1+1	0	2	0	0	0	0	0	0	
Fiber counts per tube	G.657A1 200um	6			12								24							
Central strength number	Material	FRP																		
	Diameter(mm)	1.0			1.2			2.0	2.4	1.8	2.4	1.8	2.8	2.4	2.8	1.8	3.4			
	Diameter of PE layer	/			/			/	3.6	/	/	/	/	3.4	5.0	/	/			
Outer sheath	Material	HDPE																		
	Color	Black																		
	Thickness (mm)	Approx.0.45																		
Cable diameter (± 0.3 mm)		4.0			4.5			5.3	6.9	7.5	8.0	6.1	7.2	7.7	9.3	9.6	11.2			
Cable weight (kg/km) Approx.		16			18			26	43	48	58	32	48	58	80	78	110			
For micro-duct inside (mm)		5.5~8			6~8			8~10	10~14				8~14	10~14		12~14		14~20		
Max. tensile strength (N)		200						800	1000	800	1000	1000						1500		
Crush (N/100mm)		Short term: 500 Long term: 200																		

CONTACT & INFORMATION:

Tel: (920) 328-1020 Email: info@raddnetwork.com Website: RaDDNetwork.com



Mechanical, Electrical and Environmental Test Characteristics

The finished cables can be subjected to the following mechanical, electrical and environmental conditions.

Item	Test Method	Requirements
Tensile performance	IEC 60794-1-21-E1 Load: according to short term tensile described in 3.2.2 Cable length under tension: Not less than 50m. Duration of load sustain: 1min. Velocity of transfer device: 10mm/min	The maximum fiber strain less than 0.6% under maximum tensile short term load. The maximum increase in attenuation less than 0.1dB. No change in attenuation after test at 1550nm. Under visual examination without magnification, no damage to the sheath or to the cable elements after test.
Crush	IEC 60794-1-21-E3 Load: 500N Duration of load: 1min	No change in attenuation after test at 1550nm. Under visual examination without magnification, no damage to the sheath or to the cable elements. The imprint of the striking surface on the sheath is not considered mechanical damage.
Bend	IEC 60794-1-21-E11A Mandrel radius: 10 times cable diameter Turns:10 Cycles:5	No change in attenuation at 1550nm after test. Under visual examination without magnification, no damage to the sheath or to the cable elements.
Repeated bending	IEC 60794-1-21-E6 Bending radius: 20 times cable diameter Cycles: 25 Load: 25N Duration of cycle: Approx. 2s.	No change in attenuation at 1550nm after test. Under visual examination without magnification, no damage to the sheath or to the cable elements.
Torsion	IEC 60794-1-21-E7 Cycles: 5 Length under test: 1 m Turns: $\pm 180^\circ$ Load: 40 N	The variation on attenuation for each fiber less than 0.05 dB at 1550 nm. Under visual examination without magnification, no damage to the sheath or to the cable elements. No permanent change in attenuation after test.
Temperature cycling	IEC 60794-1-22-F1 Sample length: at least 1000 m Temperature range: -30°C ~+70°C Cycles: 2 Temperature cycling test dwell time: 12 hours	The change in attenuation coefficient shall be less than 0.15 dB/km.
Water Penetration	IEC 60794-1-22-F5B Time : 24 hours Sample length : 3 m Water height : 1 m	No water leakage

Mechanical, Electrical and Environmental Test Characteristics (cont.)

Item	Test Method	Requirements
Compound flow	IEC 60794-1-21-E14 Temperature: 70°C Sample count: 5 Sample length: 200 ± 5 mm, Remove length: 100 ± 2.5 mm, Time: 24 h	No filling compound dripped.
Other parameters	According to IEC 60794 ,YD/T1460.4-2006	

Remark: “No attenuation changes” is considered as the attenuation changes ≤ 0.05 dB.

Packaging and Drum

Cable Sheath Marking

Unless otherwise specified, the cable sheath marking shall be as follows:

- Type: laser
- Contents: RaDD, the year of manufacture, the type of cable, length marking
- Interval: 1 m

Reel Length

Standard reel length: 2/3 km/reel, other lengths are also available.

Cable Drum

The cables are packed in fumigated wooden drums.

Cable Packing

Both ends of the cable will be sealed with suitable plastic caps to prevent the entry of moisture during shipping, handling and storage. The inner end is available for testing.

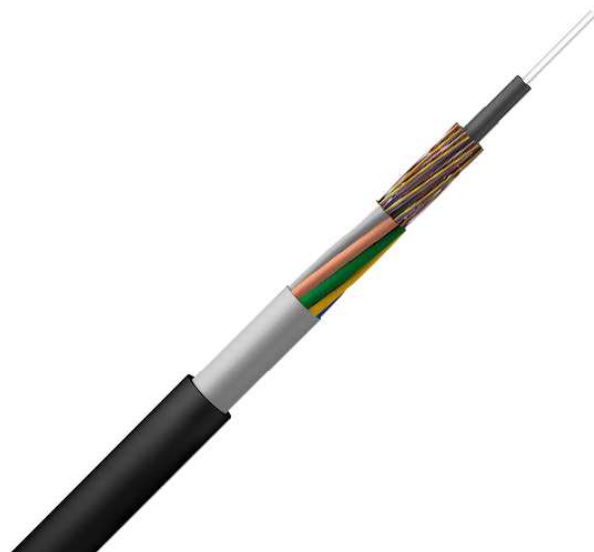
CONTACT & INFORMATION:

Tel: (920) 328-1020 Email: info@raddnetwork.com Website: RaDDNetwork.com



Loose Tube Single Jacket/Dielectric Cable

864F



Rendering is for Schematic Purposes Only

Part Number:

10-18040.01A - 864F Loose Tube SJ/Dielectric Cable

CONTACT & INFORMATION:

Tel: (920) 328-1020 Email: sales@raddnetwork.com Website: RaDDNetwork.com



General / Scope

This Specification covers the design requirements and performance standard for the supply of optical fiber cable in the industry. RaDD ensures a stable quality control system for our cable products through several programs including ISO 9001, ISO 14001 and OHS.

Cable type	Application
864F Loose Tube/SJ Dielectric Cable	Duct installation

Reference

ITU-T G.657	Characteristics of a single-mode optical fiber
IEC 60794-1-1	Optical fiber cables-part 1-1: Generic specification-General
IEC 60794-1-21	Optical fiber cables- part1-2-Generic specification-Basic optical cable test procedure-Mechanical test methods
IEC 60794-1-22	Optical fiber cables- part1-2-Generic specification-Basic optical cable test procedure-Environmental test methods
IEC 60794-3	Optical fiber cables-part 3: Sectional specification-Outdoor cables
IEC 60794-3-10	Optical fiber cables-part 3-10: Outdoor cables-Family specification for duct and direct buried optical communication cables
IEC 60794-3-11	Optical fiber cables-Part 3-11: Outdoor cables-Detailed specification for duct and directly buried single-mode optical fiber telecommunication cables

Life Time

Optical fiber cables supplied in compliance with this specifications is capable to withstand the typical service condition for a period of twenty-five (25) years without detriment to the operation characteristics of the cable.

Application

Item	Value
Operation temperature	-40 C° ~ +70 C°
Storage temperature	-40 C° ~ +70 C°
Static bending radius	10 times the cable diameter
Dynamic bending radius	20 times the cable diameter

Optical Fiber

Optical Fibers supplied in this specification meet the requirements of ITU-T G.657A1

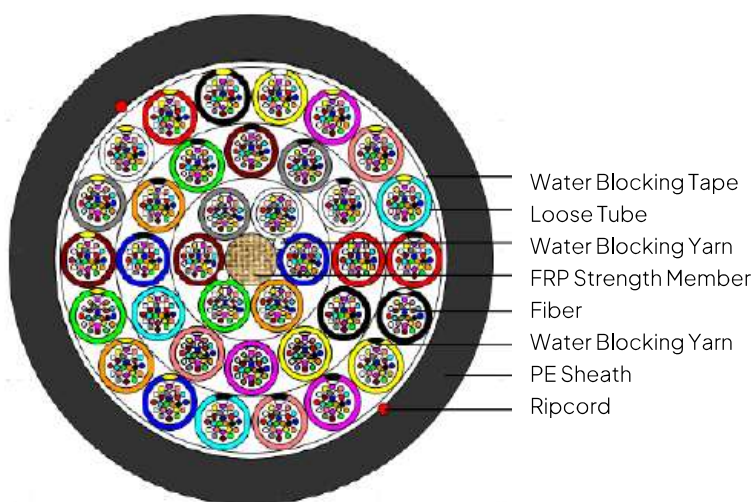
Parameter	Specification
MFD (1310nm)	8.4~9.2um
Cladding diameter	125±0.7um
Fiber diameter	235~255um, with UV coating, and colored to : 250±15um
Core/cladding concentricity error	≤ 0.5um
Coating/cladding concentricity error	≤ 12.0um
Cladding non circularity	≤ 0.7%
Cut off wavelength	$\lambda_{cc} \leq 1260\text{nm}$
Attenuation coefficient	1310nm: 0.36dB/km max after cabling 1550nm: 0.22dB/km max after cabling
Bending-loss performance of optical fibers @1550nm	≤0.25dB (10 turns around a mandrel of 30mm diameter)
Polarization mode dispersion maximum individual fibre	≤0.2ps/√km
Polarization mode dispersion link value	km√/ps 1.0≤
Zero-dispersion wavelength	1300~1324nm
Zero-dispersion slope	≤0.092ps/nm ² ·km

Optical Cable Technical Characteristics

- The unique second coating and stranding technology provide the fibers with enough space and bending endurance, which ensure good optical property of the fibers in the cable
- Accurate process control ensures good mechanical and temperature performance
- High quality raw material guarantees the long service life of cable

Cross Section of Cable

10-18040.01A



Schematic for reference only

Fiber and Loose Tube Identification

The color code of fibers and loose tube will be identification in accordance with the following color sequence, other sequence also is available.

Fiber Color Code	1	2	3	4	5	6
		Orange	Green	Brown	Grey	White
	7	8	9	10	11	12
	Red		Yellow	Purple	Pink	Aqua
	1	2	3	4	5	6
		Orange with black ring	Green with black ring	Brown with black ring	Grey with black ring	White with black ring
	7	8	9	10	11	12
	Red with black ring	Natural with black ring	Yellow with black ring	Purple with black ring	Pink with black ring	Aqua with black ring

Tube Color Code	Inner layer 1	Inner layer 2	Inner layer 3	Inner layer 4	Inner layer 5	Inner layer 6
		Orange	Green	Brown	Grey	White
	Middle layer 1	Middle layer 2	Middle layer 3	Middle layer 4	Middle layer 5	Middle layer 6
	Red		Yellow	Purple	Pink	Aqua
	Middle layer 7	Middle layer 8	Middle layer 9	Middle layer 10	Middle layer 11	Middle layer 12
		Orange with black stripe	Green with black stripe	Brown with black stripe	Grey with black stripe	White with black stripe
	Outer layer 1	Outer layer 2	Outer layer 3	Outer layer 4	Outer layer 5	Outer layer 6
	Red with black stripe		Yellow with black stripe	Purple with black stripe	Pink with black stripe	Aqua with black stripe
	Outer layer 7	Outer layer 8	Outer layer 9	Outer layer 10	Outer layer 11	Outer layer 12
		Orange with yellow stripe	Green with yellow stripe	Brown with yellow stripe	Grey with yellow stripe	White with yellow stripe
	Outer layer 13	Outer layer 14	Outer layer 15	Outer layer 16	Outer layer 17	Outer layer 18
	Red with yellow stripe		Yellow with white stripe	Purple with yellow stripe	Pink with yellow stripe	Aqua with yellow stripe

Dimensions and Descriptions

The standard optical cable structure is shown in the following table, other structure and fiber count are also available according to customer requirements.

Item	Contents	Value
		864
Loose tube	Number	36
	Outer diameter (mm)	2.9
Filler	Number	0
Max. fiber counts per tube		24
Central strength member	Material	FRP
	Diameter (mm)	3.2
	PE layer diameter (mm)	/
Water Blocking Material	Material	Water Blocking Tape(one side)& Yarn
Sheath	Material	MDPE
	Color	Black
	Thickness (mm)	Nominal: 1.6
Ripcord	Number	2
	Color	Red
Cable diameter(mm) Approx.		24.2
Cable weight(kg/km) Approx.		340

Main Mechanical and Environmental Performance

Main mechanical performance

Fiber counts	Tensile performance(N)	Crush(N/100mm)
864	2000	1500

Mechanical, Physical and Environmental Test Characteristics

The mechanical and environmental performance of the cable are in accordance with the following table. Unless otherwise specified, all attenuation measurements required in this section shall be performed at 1550nm.

Item	Test Method	Requirements
Tension	<u>IEC 60794-1-21-E1</u> Load: According to 3.2.3 Sample length: Not less than 50m. Duration time: 1min.	Additional attenuation: ≤ 0.1 dB after test No damage to outer jacket and inner elements
Crush	<u>IEC 60794-1-21-E3</u> Load: According to 3.2.3 Duration of load: 1min	Additional attenuation: ≤ 0.1 dB after test No damage to outer jacket and inner elements
Impact	<u>IEC 60794-1-21-E4</u> Radius: 300mm Impact energy: 10J Impact number: 1 Impact points: 3	Additional attenuation: ≤ 0.1 dB No damage to outer jacket and inner elements
Bend	<u>IEC 60794-1-21-E11A</u> Mandrel radius: 10*D Turns: 10 Cycles: 5	Additional attenuation: ≤ 0.1 dB No damage to outer jacket and inner elements
Repeated bending	<u>IEC 60794-1-21-E6</u> Bending radius: 20*D Cycles: 25 Load: 150N	Additional attenuation: ≤ 0.1 dB No damage to outer jacket and inner elements
Torsion	<u>IEC 60794-1-21-E7</u> Cycles: 10 Length under test: 1m Turns: $\pm 90^\circ$ Load: 150N	Additional attenuation: ≤ 0.1 dB No damage to outer jacket and inner elements
Temperature cycling	<u>IEC 60794-1-2-F1</u> Sample length: at least 1000m Temperature range: $-40^\circ\text{C} \rightarrow +70^\circ\text{C}$ Cycles: 2 Temperature cycling test dwell time: 12 hours	The change in attenuation coefficient shall be less than 0.1 dB/km
Water Penetration	<u>IEC 60794-1-21-F5B</u> Time : 24 hours Sample length : 3m Water height : 1m	No water leakage.

Other parameters According to IEC 60794-1

Packaging and Drum

Cable Sheath Marking

- Unless otherwise specified, the cable sheath marking shall be as follows:
- Color: white
- Contents: RaDD, the year of manufacture, the type of cable, cable number, length marking Interval: 3 ft
- Outer sheath marking legend can be changed according to user's requests.

Reel Length

Standard reel length: 10,000 ft/reel, other length is also available.

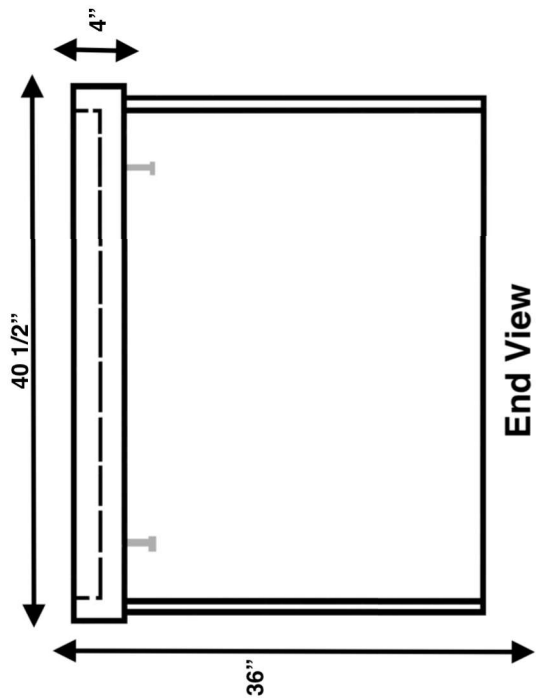
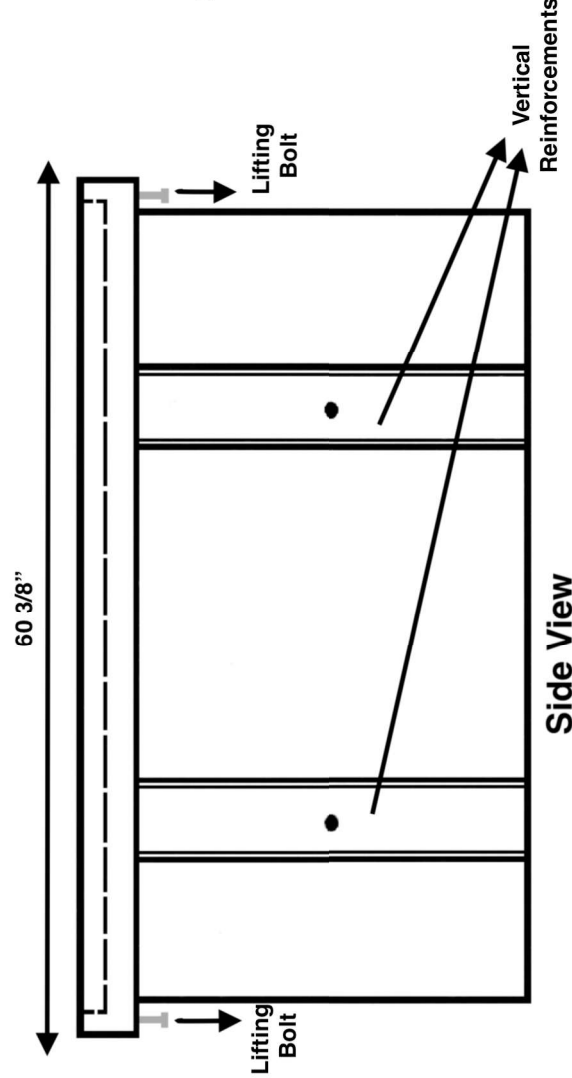
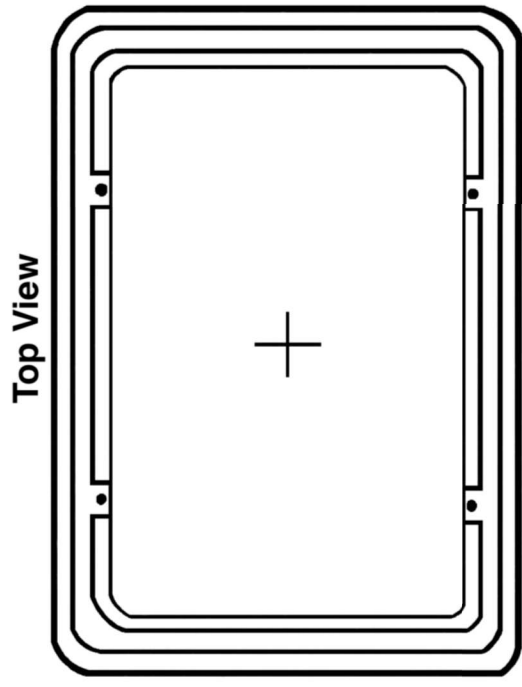
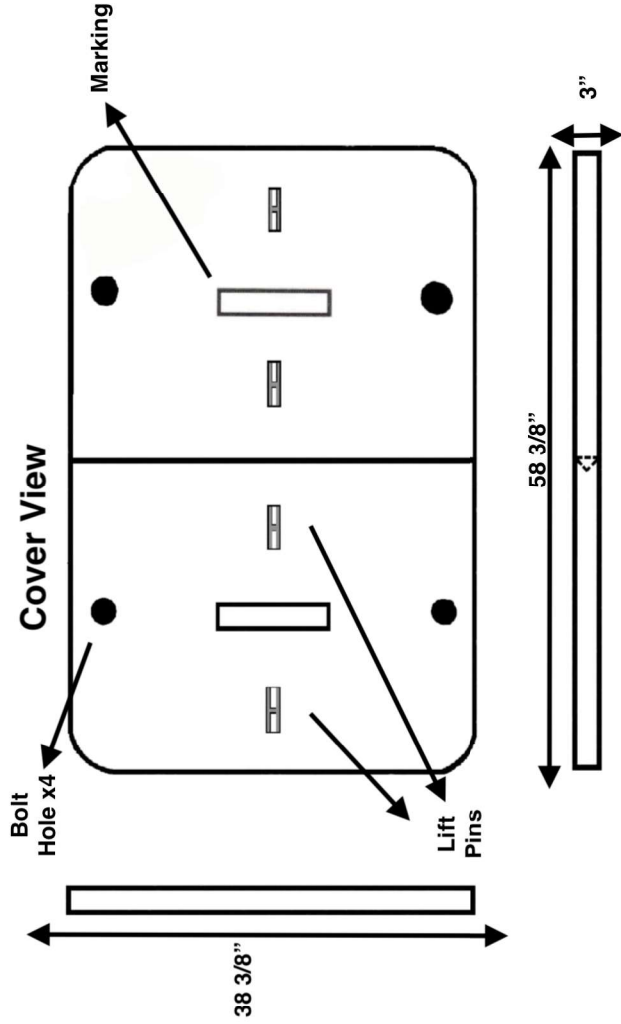
Cable Drum

The cables are packed in fumigated wooden drums.

Cable Packing

Both ends of the cable will be sealed with suitable plastic caps to prevent the entry of moisture during shipping, handling and storage. The inner end is available for testing.

TV366036 Polymer Concrete Box & Cover (2 Piece Lid)



Loading Requirements
Vertical
Design Load: 22,500 lbs
Test Load: 33,750 lbs

Lateral
Design Load: 800 lbs/sq. ft.
Test Load: 1,200 lbs/sq. ft.

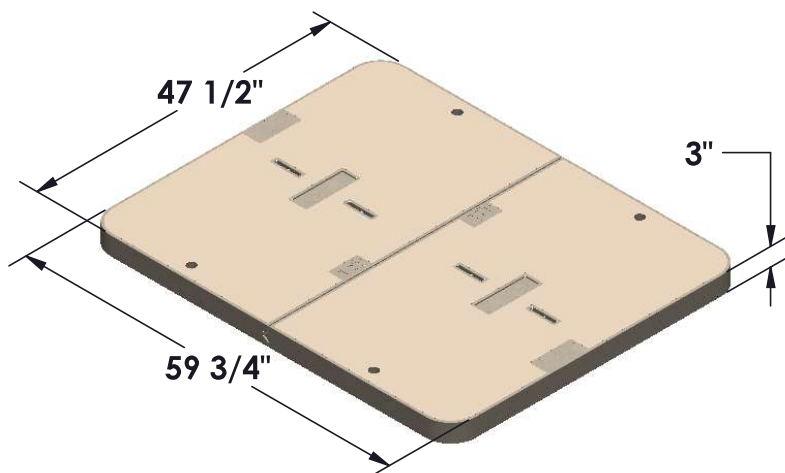
Part No. TV366036PCT22	
Body: Polymer Concrete	
Cover: Polymer Concrete	
Weight: 970LB	Tolerance: +/- 1/8"
Rating: Tier 22	



POLYMER CONCRETE BOX & COVER



MODEL 4860/48 PC



COVER

Dimensions* 59 3/4" x 47 1/2" x 3"
Material Polymer Concrete
Style Flush Split
Fasteners 3/8-16 x 3" SS Hex Bolt
Lift points 1/4" SS Rod

Non-skid cover

Custom Logo Available

Body

Dimensions* 62 1/2" x 50 1/8" x 48"
Material Polymer Concrete
Lift Points 4-SS Lifting Bolts

Additional body depths available

Description

ANSI/SCTE 77 Approved Tier 22

*Vault Weight: 992 lbs

*Cover Weight: 198 lbs each (2)

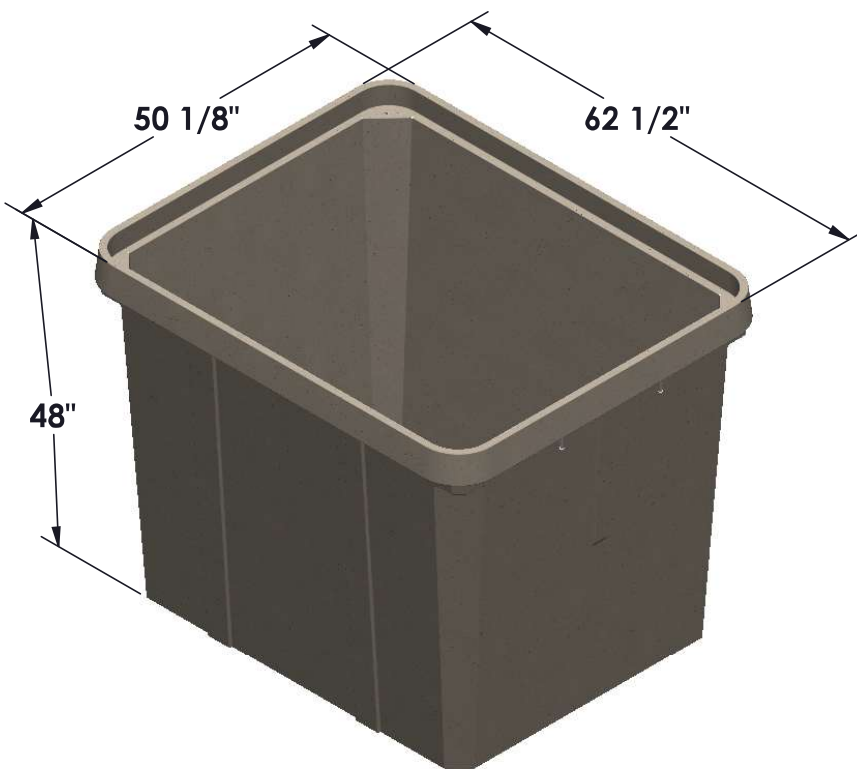
*Total Assembly weight: 1,388 lbs

Vaults per pallet: 1

Vaults per truck: 12

*Truckload weight includes
pallets: 17,136 lbs

*All weights and dimensions are approximate



Vertical Loading Requirements

Design Load: 22,500 lbs

Test Load: 33,750 lbs

Lateral Loading Requirements

Design Load: 800 lbs/sq. ft

Test Load: 1200 lbs/sq. ft

Optional Accessories

EMS Frequency Markers

Metallic Locate Plates

Mouse Holes

Conduit Ports

Cable Racking Hooks



FOSC™450 D6 Fiber Optic Splice Closure, Gel Cable sealing, no pre-installed tray, 6 cable attach., six ground feedthrough lugs, with test valve

- Single-ended, O-ring sealed dome closure for splicing feeder and distribution cables
- Gel cable sealing technology allows easy adding or removing of a wide size range of cables
- Compatible with most common cable types: e.g. loose tube, central core, ribbon fiber
- FOSC splice trays hinged for access to any splice without disturbing other trays
- Closure can be used in aerial, pedestal and underground (up to 5 meters) environments
- Compatible with CommScope's CWDM modules and optical splitter trays

Product Classification

Product Type	Single-ended, round fiber closure
Regional Availability	Latin America North America
Product Series	FOSC 450
Product Brand	FOSC™

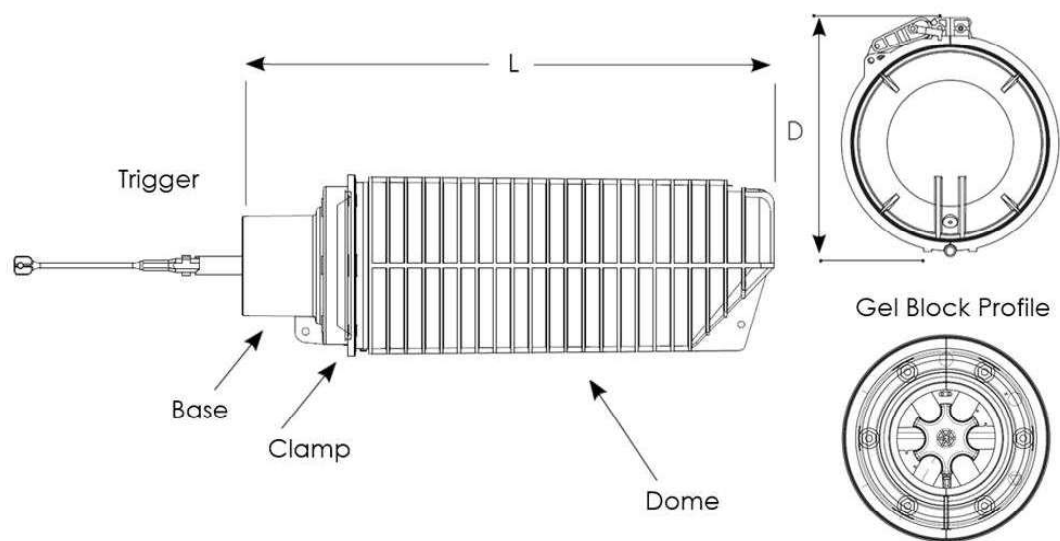
General Specifications

Splicing Type, Supported	Mass fusion Single fusion
Splicing Capacity, Mass Fusion, maximum	1152
Splicing Capacity, Single Fusion, maximum	576
Color	Black
Includes	Cable termination attachments (6)
Network Area Type	Feeder
Package, quantity	1
Packaging Type	Box Carton
Splice Tray Included, quantity	0
Splicing Capacity, Single Splice, 12 fibers, maximum	576
Splicing Capacity, Single Splice, 6 fibers, maximum	288

Dimensions

Main Cable Diameter, maximum	25.00 mm
Diameter	254.00 mm 10.00 in
Diameter, with clamp	292 mm 11 in
Length	753.00 mm 29.65 in
Weight	9.07 kg 20.00 lb

Dimension Drawing



Mechanical Specifications

Cable Sealing Type	Compressed gel
Cable Entry Drop Port Style	Round
Cable Entry Main Port Style	Round
Cable Ports, quantity	6
Closure Sealing Type	Dome-to-base clamp with O-ring
Closure Style	Single-ended
Fiber Storage Basket Type	D6 standard size
Ground Feed-through Lugs, quantity	6
Material Type	Impact-resistant polymer
Mount Type	Pole Strand Wall

Environmental Specifications

Environmental Space	Below ground Buried
Qualification Standard	IEC 61300, 5 m waterhead
Water Resistance	Flash test valve at 5 psi (40 kPa)

Regulatory Compliance/Certifications

Agency	Classification
RoHS 2011/65/EU	Compliant
ISO 9001:2015	Designed, manufactured and/or distributed under this quality management system





FOSC650 Fiber Optic Splice Closure, no tray, no gel sealing, includes standard basket, 12 grounding lugs and valve, Sealing and cover for unused ports ordered separately.

- Single-ended base with four-latches dome design and reliable cable attachment system
- There are 3 cable segments: 2 for use with the small gel seal blocks/entrance ports and 1 for use with the large gel seal blocks/entrance ports
- Fully removable base for easy cable access during installation and provisioning
- Superior gel sealing technology incorporated in interchangeable and fully wrap around gel segments
- Compatible with a wide range of fiber cable sizes and types

Product Classification

Regional Availability	Latin America North America
Product Type	Single-ended, square fiber closure
Product Brand	FOSC®

General Specifications

Cable Entry Drop Port Style	Gel block
Cable Entry Main Port Style	Gel block
Cable Ports Quantity, total	3 multi-out ports (12 cables)
Cable Sealing Type	Compressed gel
Closure Sealing Type	Over-centered latches with rubber gasket
Closure Style	Single-ended
Color	Black
Flash Test Valve	Pre-installed
Ground Feed-through	None
Mounting	Pole Strand Wall
Network Area Type	Feeder Trunk
Splice Tray Included, quantity	0
Splice Tray Type Included	No trays
Splicing Capacity, Mass Fusion, maximum	6912
Splicing Capacity, Single Fusion, maximum	3456
Splicing Capacity, Single Splice, 12 fibers, maximum	3456

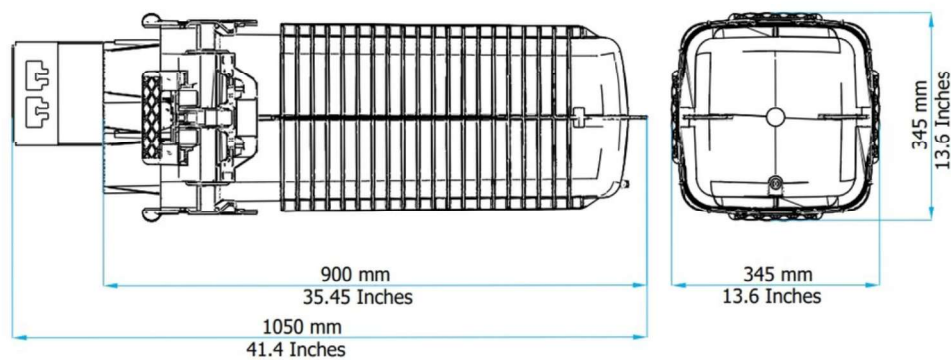
760251859 | FOSC650-NNN-NT-0-S12V

Splicing Capacity, Single Splice, 6 fibers, maximum	180
Splicing Type, Supported	Mass fusion Single fusion

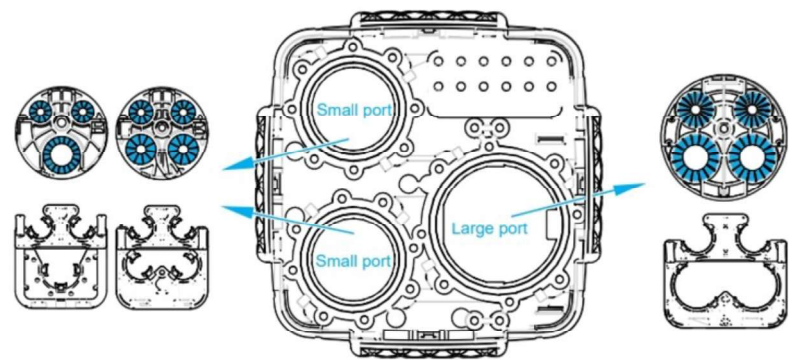
Dimensions

Height	345.44 mm 13.6 in
Width	345.44 mm 13.6 in
Length	900.43 mm 35.45 in
Length, with trigger	1,050.29 mm 41.35 in
Main Cable Diameter, maximum	40.64 mm 1.6 in

Dimension Drawing



Port Configuration



	Upper -Small Ports-		Lower -Large Ports-	
Gel Seals	# Ports	Cable Outside Diameter Min. - Max.	# Ports	Cable Outside Diameter Min. - Max.
Small 3-way 	2	8 - 20 mm 0.32 - 0.79 inch	1	25 - 35 mm 1.0 - 1.38 inches
Small 4-way 	2	8 - 22 mm 0.32 - 0.87 inch	2	10 - 28 mm 0.39 - 1.1 inches
Large 4-way 	2	18 - 25 mm 0.7 - 1.0 inch	2	25 - 40 mm 1.0 - 1.6 inches

Material Specifications

Material Type

Rugged polymer

Environmental Specifications

Environmental Space

Below ground | Buried

Qualification Standards

IEC 60529, IP68 + 5 m waterhead

Packaging and Weights

Packaging quantity

1

Packaging Type

Box | Carton

Weight, net

11.79 kg | 25.992 lb

Regulatory Compliance/Certifications

Agency	Classification
ISO 9001:2015	Designed, manufactured and/or distributed under this quality management system